

Question – 1

1. Social Connections

A popular social media platform provides a feature to connect people online. Connections are represented as an undirected graph where a user can see the profiles of those they are connected to.

There are $connection_nodes$ users numbered 1 to $connection_nodes$, and $connection_edges$ connections where the i^{th} pair connects nodes $connection_from[i]$ and $connection_to[i]$. The $queries$ array contains node numbers. Find the number of users whose profiles are visible to $query[i]$. Report an array of integers where the i^{th} value is the answer to the i^{th} query.

Example

```
connection_nodes = 7
connection_edges = 4
connection_from = [1, 2, 3, 5]
connection_to = [2, 3, 4, 6]
queries = [1, 3, 5, 7]
```

```
graph LR
    1 --- 2
    2 --- 3
    3 --- 4
    5 --- 6
    7
```

Test Results **Custom Input**

Sample Output 0

```
4
4
1
```

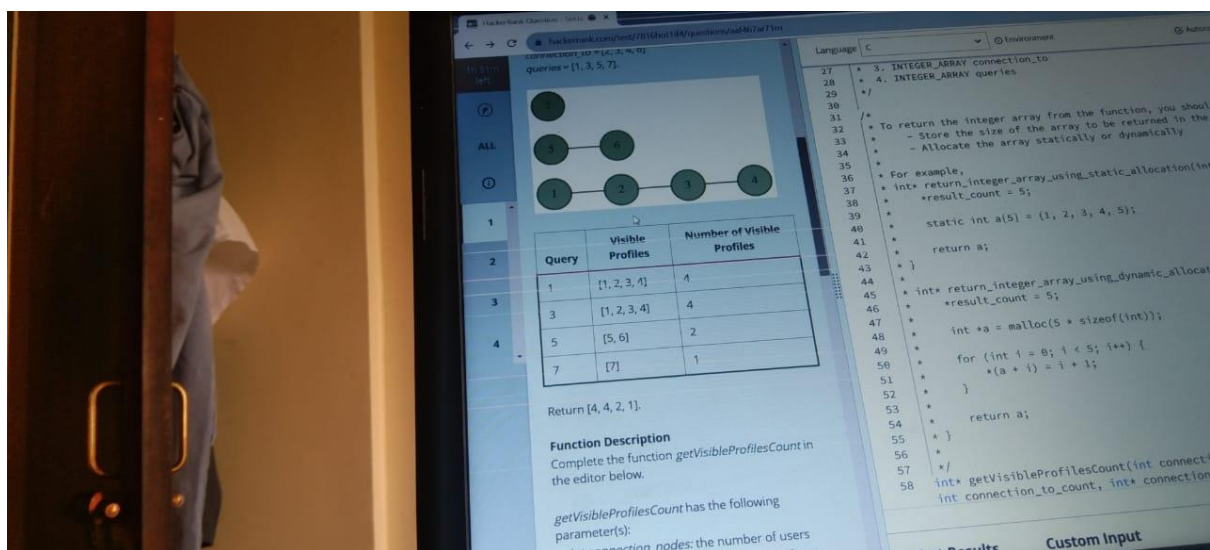
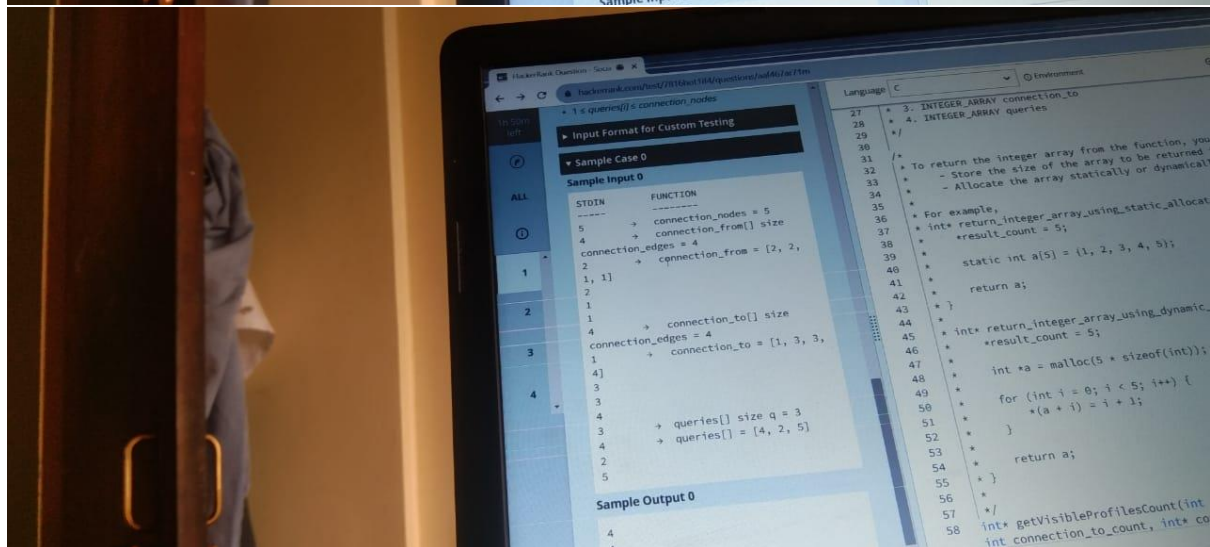
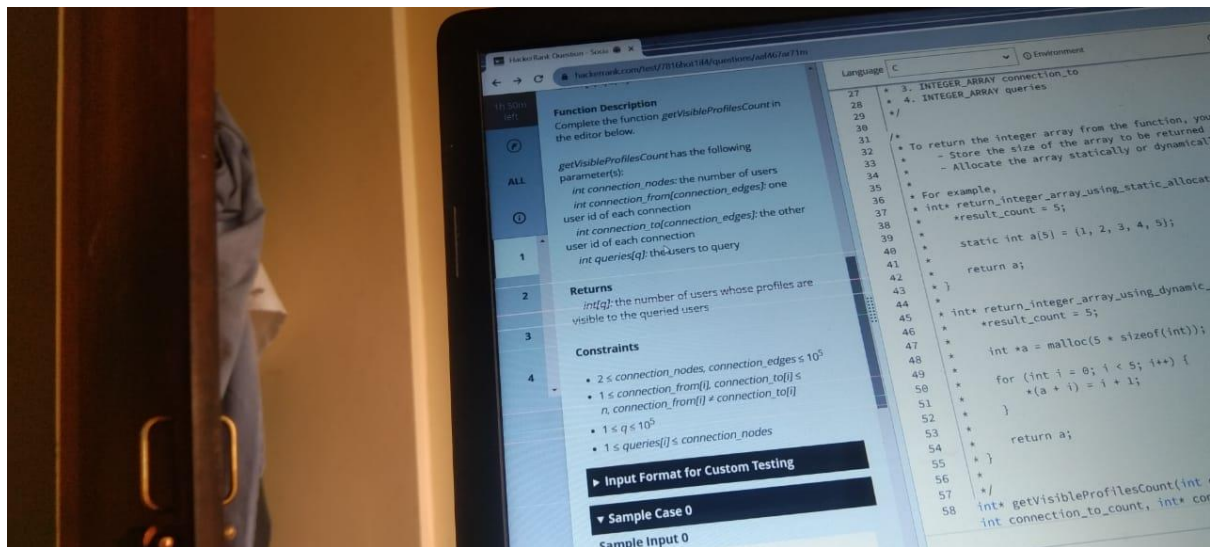
Explanation

```
graph LR
    1 --- 2
    1 --- 3
    1 --- 4
    5
```

Each of the users can see the profiles of 4 users except for 5 who can see only its own profile.

Sample Case 1

Test Results **Custom Input**



Question -2

2. Bits in Archie

Alan Emtage is a computer scientist from Barbados who is widely known for implementing the world's first internet search engine, Archie. While implementing Archie, Emtage needs to test some bitwise functions required to make the search engine compute more efficiently.

Emtage takes an array of positive integers and a threshold value. The goal is to count the number of unordered pairs of elements in the array such that the sum of the number of set bits in the binary representation of their bitwise OR and bitwise AND is greater than or equal to the given threshold value.

Example

Consider Emtage has the array, $arr = [2, 4, 6, 8, 10]$, and the threshold value $k = 4$.

Let $|$ denote the OR operation, and $\&$ denote the AND operation, and $/x/$ denotes the number of set bits in the binary representation of integer x .

There are 10 unordered pairs in Emtage's array:

- For pair (2, 4), $|2|4| + |2\&4| = 2$, which is less than threshold 4.

Function Description

Complete the function `countBit` in the editor below.

`countBit` has the following parameter(s):

- `arr[0]...arr[n-1]`: an array of n positive integers
- `k`: the threshold value

```
1 // Complete the 'countBit' function below.
19
20 /*
21  * Complete the 'countBit' function below.
22  * The function is expected to return a LONG_INTEGER.
23  * The function accepts following parameters:
24  * 1. INTEGER_ARRAY arr
25  * 2. INTEGER k
26  */
27
28 long countBit(int arr_count, int* arr, int k) {
29
30 }
31
32
33 > int main()
```

There are 10 unordered pairs in Emtage's array:

- For pair (2, 4), $|2|4| + |2\&4| = 2$, which is less than threshold 4.
- For pair (2, 6), $|2|6| + |2\&6| = 3$, which is less than threshold 4.
- For pair (2, 8), $|2|8| + |2\&8| = 2$, which is less than threshold 4.
- For pair (2, 10), $|2|10| + |2\&10| = 3$, which is less than threshold 4.
- For pair (4, 6), $|4|6| + |4\&6| = 3$, which is less than threshold 4.
- For pair (4, 8), $|4|8| + |4\&8| = 2$, which is less than threshold 4.
- For pair (4, 10), $|4|10| + |4\&10| = 3$, which is less than threshold 4.
- For pair (6, 8), $|6|8| + |6\&8| = 3$, which is less than threshold 4.
- For pair (6, 10), $|6|10| + |6\&10| = 4$, which is greater than or equal to threshold 4.
- For pair (8, 10), $|8|10| + |8\&10| = 3$, which is less than threshold 4.

Since there is only one pair, (6,10), which meets the required condition, the answer is 1.

Function Description

Complete the function `countBit` in the editor below.

`countBit` has the following parameter(s):

- `arr[0]...arr[n-1]`: an array of n positive integers
- `k`: the threshold value

```
19
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28 long countBit(int arr_count, int* arr, int k) {
29
30 }
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32
33 > int main()
```

Returns

`int`: an integer denoting the number of unordered pairs of array elements such that the sum of the number of set bits in the binary representation of their bitwise OR and bitwise AND is greater than or equal to the threshold value.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq arr[i] \leq 10^9$
- $1 \leq k \leq 100$

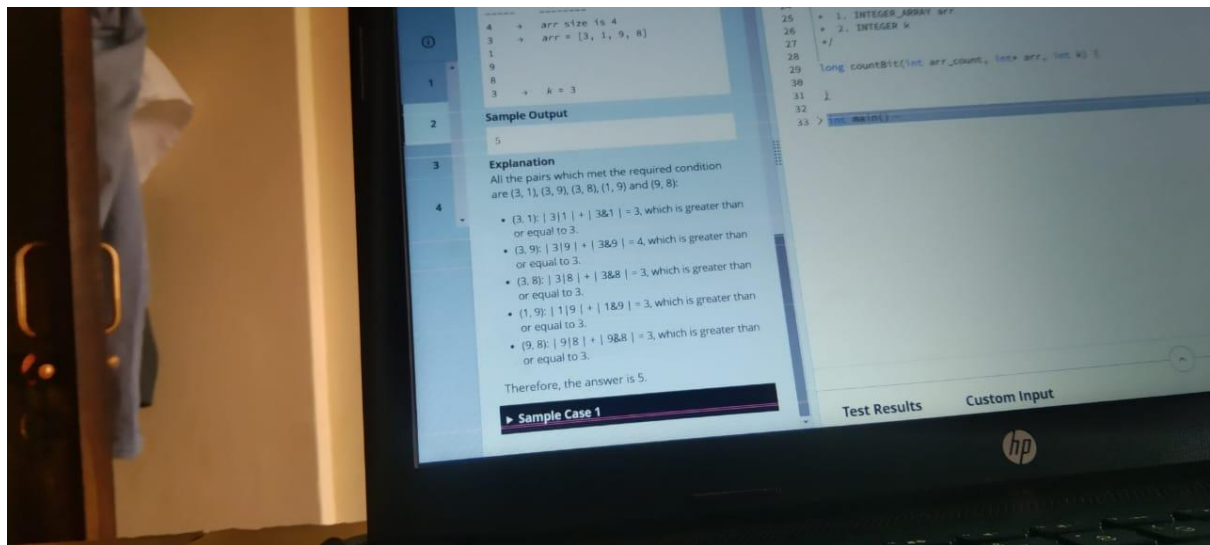
Input Format For Custom Testing

Sample Case 0

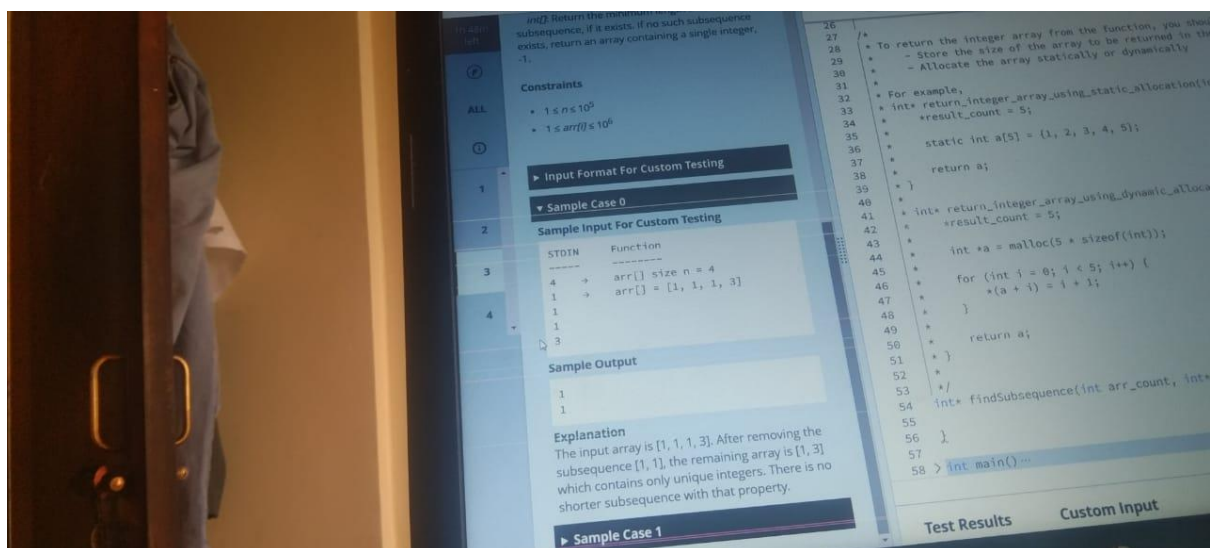
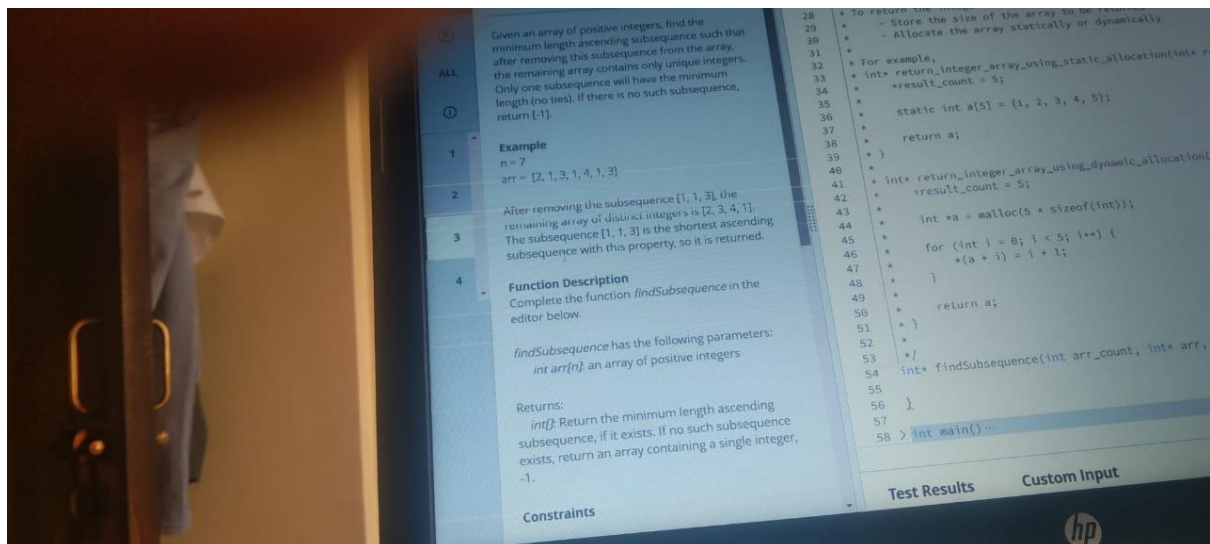
Sample Input For Custom Testing

STDIN	FUNCTION
4	$\rightarrow$ arr size is 4
3	$\rightarrow$ arr = [3, 1, 9, 8]
1	
9	

```
23
24 * The function is expected to return a LONG_INTEGER.
25 * The function accepts following parameters:
26 * 1. INTEGER_ARRAY arr
27 * 2. INTEGER k
28
29 long countBit(int arr_count, int* arr, int k) {
30
31 }
32
33 > int main()
```

Question3



Option – 4

